Agua a Tiempo MX

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The problem in my words

When the water supply stops, households often do not know whether a notice is reliable, who is responsible for updating it, or when they will hear more. My household experience includes approximately four street-wide cuts a year, insufficient stored water and a reported MX$3,000 bimonthly bill. This is one household's experience, not a population estimate. Residents should not have to buy another membership to receive basic information about a service they already pay for.

My declared slice

Build an incident-notice review page, an SMS preview and a pilot cost/payment scorecard. This implements Valeria's declaration in the team Blueprint, particularly conditions 2, 4 and 6, while preserving every access and privacy condition. It does not replace Regina's private reserve ledger or Asael's full operational workflow.

Exact user and beneficiary

Primary user: a designated pilot coordinator reviewing utility-sourced incidents and checking notice quality and costs. **Beneficiaries:** residents in the affected service zone, including people without smartphones or water storage. A resident need not use this dashboard to receive a basic notice.

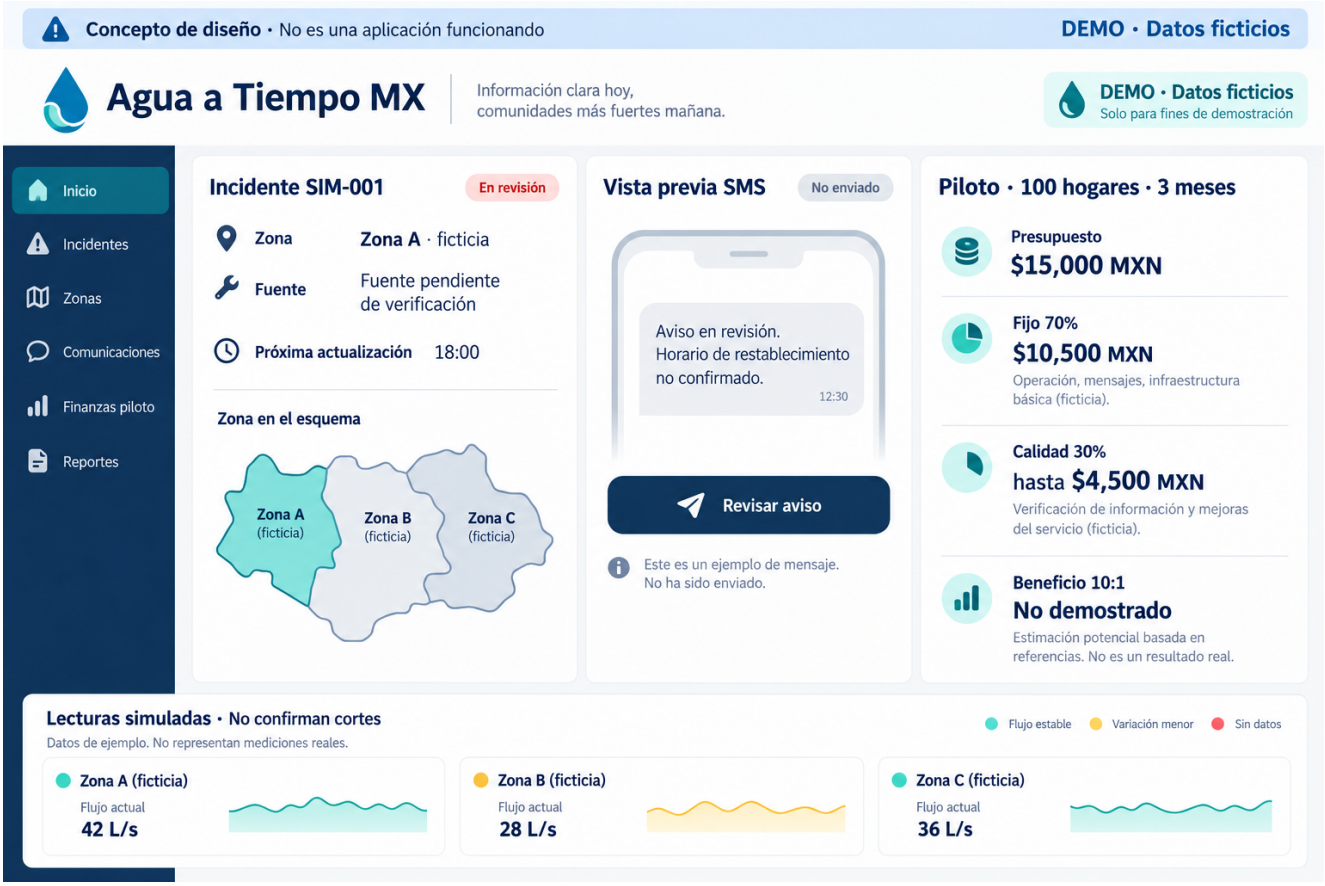
Success before the module closes

Before the module closes, a reviewer can load a fictional incident, identify missing or stale evidence, review an ML-assisted flag, simulate an authorized-human review, generate a clearly unsent SMS preview and see reproducible cost and quality calculations at a public URL. A second deployment must incorporate a documented mechanical bug fix; the worst observed persona confusion must also be fixed and retested.

Planning status: only folder creation and environment checks have occurred. This Packet precedes application implementation. No deployment, real pilot, confirmed utility partnership, SMS delivery or measured benefit is claimed.

The review workspace

The image below is an AI-generated design concept, not evidence of a working application. The implementation must preserve the labels even if individual visual details change.



Critical interface states

Pending: no verified source or authorized review; official-ready preview blocked. **Reviewed in demo:** human-role simulation only; never a real official confirmation. **Stale:** next update overdue or source older than the configurable 12-hour demo threshold; require a new review. **Corrected:** retain previous text, correction reason and new timestamp.

Each preview shows incident ID, fictitious zone, source status, responsible role and next update. Unknown restoration time remains unknown. The screen always says “DEMO - Datos ficticios” and “Vista previa SMS - No enviado.”

Evidence before a notice

Feature flow

Step	Decision and resulting behavior
1. Load input	Load a fictional incident and dated, explicitly simulated reserve readings. Zones are fictitious polygons. Readings never confirm a public-network event.
2. Validate	Check required fields, allowed zones, text lengths, dates and costs. Reject invalid input with an actionable message.
3. Review evidence	Deterministic checks identify missing source, responsible role, next update or stale data. ML adds an advisory flag; it cannot override blocking rules.
4. Human review	A simulated authorized reviewer accepts or returns the notice. The UI states that this is a role simulation, not real authorization.
5. Preview and log	Generate SMS text and offline-contact script; record simulated dispatch/receipt states separately. Correction reopens review.
6. Evaluate	Compute notice timing, sampled comprehension, preparation and full costs with denominators; no eligible real events means inconclusive.

Swimlane: who does what

Actor / lane	Responsibility
Utility contact	Outside the demo: accept verification, correction and escalation responsibility before field spending; supply evidence and updates.
Coordinator	Review source and dates, inspect warnings, simulate human approval and review corrections/costs.
System / ML	Validate, flag uncertainty, match zones, generate unsent previews and calculate metrics. Never confirm supply events.
Resident / offline contact	Outside the demo: receive SMS or agreed offline contact, ask for correction and optionally answer comprehension/preparation checks. No app required.

The Mermaid source for this flow and swimlane is included on page 8 and must be copied into docs/PACKET.md before the first application-code commit.

Adapt the rails locally

Benchmark line

Selected leading benchmark: PUB Singapore's Smart Water Meter / MySmartWaterMeter system provides household consumption visibility and suspected-leak alerts. It is a relevant reference, not a demonstrated universal winner or a direct substitute for incident-notice review.

How mine localizes: Agua a Tiempo MX focuses on Spanish-language, human-reviewed supply-incident notices, SMS/offline access and transparent pilot payment evidence without requiring household hardware or charging residents.

Primary sources: PUB “What is Smart Water Meter” and “Help & Feedback: Report.” PUB's reporting guidance explicitly asks users to check interruption notices by letter or SMS. These sources support the design benchmark, not Mexican pilot outcomes.

Three-year light charter

If this slice works, Agua a Tiempo MX becomes a utility-funded preparation and accountability service connecting private household reserve records with verified incident updates. It integrates approved utility sources, optional sensors and consented aggregate evaluation while preserving SMS and offline access for basic notices. Expansion depends on real-event evidence, an accepted operational owner and full costs, rather than invented savings or promises to restore water.

Scope cut

We are not building physical sensors, repair services, a water marketplace, pooled repair insurance, payment collection, real SMS delivery, a production utility integration or public household maps. We do not predict causes, restoration times or water quality. We do not collect real names, phone numbers, addresses or household readings in this public demo.

Sources and design authority

1. Team Blueprint: BLUEPRINT_team_Week11.pdf, supplied by the user; declarations, six conditions and pilot budget.
2. PUB: https://www.pub.gov.sg/-/media/PUB/PDF/swm/How_Smart_Water_Meter_Works.pdf
3. PUB: https://eservices.pub.gov.sg/helpandfeedback/report
Sources checked October 21, 2026. All additional thresholds and payout mechanics in this Packet are proposed demo design choices, not team-approved procurement terms.

A small, inspectable build

Layer	Choice and acceptance condition
Interface / hosting	HTML, CSS and JavaScript; GitHub repository; Vercel free-tier static hosting. Confirm current free-tier limits during setup. Responsive keyboard-accessible forms.
IoT / sensing	Local simulated feed: timestamp, fictitious zone, reserve percentage and simulation label. Include fresh, stale and missing-reading cases; basic notice access never depends on a reading.
ML	Small logistic-regression classifier trained locally in JavaScript on labeled synthetic notice examples, with fixed-seed train/held-out split. Features: source age, missing fields, notice lead time and text features. Show evaluation counts, precision/recall and limitations; no real-world accuracy claim.
Geodata	Local GeoJSON with three fictional zones and point-in-polygon matching. Include an outside-zone case. No browser location request and no household markers.
Data / privacy	Bundled fictional fixtures and in-memory state only. No personal-data storage, analytics, backend or database. Reload resets edits. Clearly instruct users to use fictional data only.
Validation / output	Allowlisted zones and event types; finite nonnegative costs; reserve 0-100; dates parsed and cross-checked; bounded text; render user text with textContent.
Evidence	docs/PACKET.md, docs/BUILD_PROMPT.md, DECISIONS.md, test logs, model evaluation and deployment records. Keep real test outcomes separate from planned cases.

Security floor

No secrets or paid APIs are needed. No personal data is stored; Supabase Auth and RLS are not applicable to this fictional in-memory demo. Any later personal-data storage requires authentication and row-level access controls before release. Validate every form, avoid HTML injection and use only labeled fictional seed data. A role-selection control is a demo interaction, not authentication or real authorization.

ML limits and release gate

Synthetic labels encode designed examples, not observed service outcomes. Keep blocking evidence checks deterministic; ML never makes official decisions or determines payment. If training/evaluation fails, show ML unavailable and document the stack requirement as unmet until fixed; do not relabel hard-coded rules as ML.

Budget without invented benefits

Blueprint pilot: 100 households, three months, total cap MX$15,000, proposed utility-funded. Buyer validation is absent. No resident fee; no field spending before the utility contact accepts verification, corrections and escalation.

Measure	Formula / proposed treatment
Fixed component	15,000 x 70% = MX$10,500. Show as proposed component, not paid or earned automatically.
Quality pool	15,000 x 30% = maximum MX$4,500. Display separately from incurred costs and payments actually recorded.
Illustrative unit budget	MX$150 per household for the three-month pilot; MX$50 per household-month. This is budget allocation, not a resident price.
Full cost	Sum entered labor, messaging, offline access, setup and other validated costs; disclose donated time and in-kind inputs. Remaining budget = 15,000 minus full cost.
10:1 evidence	Required attributable benefit = 10 x full cost. At the cap this is MX$150,000. If benefit is absent/unverified, show Not demonstrated; never infer benefit from notice counts.

Proposed quality-payment rule for the demo

Use three gates: at least 90% of eligible scheduled notices sent at least 12 hours early; at least 80% comprehension among sampled recipients; and at least 70% preparation among households able to prepare. Display each numerator and denominator. Missing observations or a zero denominator makes the affected gate inconclusive, not 0% or 100%.

For a simple reviewable proposal, show the MX$4,500 pool as eligible for review only if every gate passes with real evidence; otherwise show unmet or inconclusive. Never automatically disburse funds. Synthetic scenarios may illustrate the arithmetic but cannot establish pilot eligibility. This all-or-nothing gate is an implementation proposal; the Blueprint specifies the 70/30 split but no payout formula.

Operational evidence to retain

Scheduled start, utility input time, review time, simulated provider dispatch, sampled receipt checks, correction count/reasons and next update. Distinguish late source input from late provider action. Track unexpected failures separately; they are not eligible for the scheduled 12-hour metric. Record false alarms, excluded households and reasons, including no phone/no storage. No real event means inconclusive.

Prove, fix, redeploy

Test	Expected result
Missing / unauthorized source	Block official-ready preview; show missing evidence. Demo review cannot claim real authorization.
Fresh vs. stale feed	Label stale/missing reserve data; still permit access to basic notice information. No causal claim about the network.
Timing boundary	Exactly 12 hours qualifies; 11h59m does not. Unexpected events excluded from scheduled denominator.
Money and empty evidence	70/30 arithmetic correct; reject negative/NaN costs; over-budget state visible; zero denominators return inconclusive.
Correction / safety	Changed source or dates invalidates prior review. Unknown restoration remains unknown. HTML-like input renders as text.
ML / geodata	Report held-out evaluation and mistakes; test outside-zone points; advisory ML cannot bypass evidence rules.
Access stress cases	No-phone and no-storage cases retain notice access and show agreed offline-route requirement. No delivery or route is claimed operational.

Mechanical pass: evidence to produce

Run the planned checks against deployment 1. Record an actually observed bug with reproduction steps, expected/actual result and screenshot or console evidence. Fix it in a dedicated commit, rerun the relevant checks and deploy version 2. Do not manufacture a bug or mark a planned case as a completed test.

Persona pass: fresh-chat test

Synthetic persona: a 54-year-old resident who uses SMS, reads slowly, distrusts unfamiliar apps and has limited storage. This is a proposed test persona, not an interviewed person. In a fresh chat, provide screenshots in order and ask her to explain whether the notice is official, whether an SMS was sent, when water returns and what she can do without storage. Record every confusion, fix the worst one and repeat the affected task. Add a coordinator walkthrough for the financial page.

Commit and deploy plan

1. Packet, mockup and implementation prompt before code. 2. Validated incident review and correction state. 3. Simulated sensing, actual ML and fictitious geodata. 4. SMS/offline previews and cost/payment scorecard; deployment 1. 5. Actual mechanical bug fix; deployment 2. 6. Persona-driven change and retest, if needed. Each session ends by updating DECISIONS.md with decisions and the next first move, committing and pushing.

Submission: live URL, GitHub link, DEMO_Valeria.mp4 (3-minute walkthrough plus 30-second mind-change), PACKET_Valeria.pdf, PERSONA_Valeria.pdf and full BUILDCHAT_Valeria.pdf. Preserve actual development dialogue and deployment evidence.

Copy into docs/PACKET.md

These fenced Mermaid blocks define the feature flow and actor swimlane. GitHub renders them in the Markdown Packet; arrows and lane responsibilities are also described on page 3.

````
```mermaid
flowchart TD
  A["Fictional incident + simulated readings"] --> B["Validate inputs"]
  B --> C{"Evidence complete and current?"}
  C -->|No| D["Show gaps; return for correction"]
  D --> B
  C -->|Yes| E["ML advisory + fictitious zone match"]
  E --> F{"Simulated human review accepts?"}
  F -->|No| D
  F -->|Yes| G["Unsent SMS + offline script preview"]
  G --> H["Log demo states; compute costs and metrics"]
  H --> I["Correction invalidates review"]
  I --> B
```

```mermaid
flowchart TD
  subgraph Utility["Utility contact - real pilot prerequisite"]
    U["Accept source verification and escalation ownership"]
  end
  subgraph Coordinator["Coordinator - demo role"]
    C["Inspect source and simulate human review"]
    R["Review corrections and financial evidence"]
  end
  subgraph System["System and ML"]
    S["Validate; advisory flag; match zone"]
    P["Generate unsent SMS and offline script"]
  end
  subgraph Resident["Resident - delivery outside demo"]
    H["Receive via SMS or agreed offline route"]
    Q["Optional receipt and comprehension check"]
  end
  U --> C
  C --> S
  S --> C
  C --> P
  P -. "Real pilot only" .-> H
  H --> Q
  Q --> R
  R --> C
```
````